

RAHUL MITRA

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EDUCATION

National University of Singapore

Singapore

Bachelor of Engineering, Industrial Systems Engineering

Aug 2023 – Jul 2027

- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.
- Second Major in Computer Science; Minor in Mathematics. Coursework: Software Engineering, Artificial Intelligence, 3D Computer Vision, Simulation, Statistics, Operations Research, Probability, Computer Systems.

Anderson Serangoon Junior College

Singapore

Singapore-Cambridge GCE A-Level

Feb 2019 – Dec 2020

EXPERIENCE

STMicroelectronics

Singapore

Operations Research (NUS System Design Project), Contract

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

Amazon

Singapore

Robotics Vision Engineer (BlendED AI+X)

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.
- Engineer processing and evaluation stages that deliver perception-ready frame quality for downstream robotics vision.

Abbott Laboratories

Singapore

Operational AI Systems & Data Engineer (Contract)

Jun 2026 – Present

- Build an air-gapped process-intelligence platform for globally distributed spray-drying plants, spanning GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted quantized Gemma model.
- Reverse-engineer four plants' APC steady-state optimizers from their Excel master files into a single HiGHS linear program with a tracking QP and PLC predictor, returning setpoints, binding constraints, and shadow prices at millisecond latency without a commercial solver license.

Abbott Laboratories

Singapore

Data Analytics and Project/Product Management Intern

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling, translating production objectives and constraints into decision-ready schedules.
- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of unseen raw data with zero execution failures.
- Productionized a team-built APC simulator for live manufacturing and engineer training, packaging it in Docker and deploying it to Azure App Service.

People's Association

Singapore

Software Engineer (Citizen Developer)

Sep 2025 – Present

- Built an end-to-end digital garden-plot balloting platform spanning admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and stakeholder management across product, engineering, and operations; awarded the S\$20,000 Sparks Community Innovation Fund.

YesWeHack

Remote

Independent Bug Bounty Researcher

May 2024 – Present

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports using Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.

Republic of Singapore Navy

Singapore

Base Support Assistant

Jan 2022 – Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

PROJECTS AND COMPETITIONS

AsyncDDGS — Open-Source Async Search Client (PyPI)

2024 – Present

- Authored and maintain an asyncio-first, aiohttp-based DuckDuckGo client on PyPI with pytest CI/CD, adopted across AI tooling, Discord bots, and search projects.

OnTheSpectrum — Codex & Blender MCP 3D Pipeline	May 2026
• Built a local-first Blender-to-Three.js asset pipeline using Codex and Blender MCP, generating GLB assets, metadata, previews, playable-world QA, and Remotion video workflows.	
Arcane Security Tooling Family	Mar 2026
• Built Rust-forward security and red-team tooling (Arcane, Arcane-PP, Arcane-GLM, Arcane-OCR) for automation, OCR, and adversarial workflows.	
Volt Pulse SG — SMU Hack For Cities 2026 (Top 8 of 50+ teams)	Jan 2026
• Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.	
Maritime Deficiency Severity Forecasting — Maritime Hackathon 2025 (Team Lead)	2025
• Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers, gradient clipping, and hyperparameter optimization on the ASPIRE 2A supercomputer.	
AgeWellLah.AI — NUS HealthHack 2025	2025
• Built an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.	
SmartExam — GovTech x NTUPC Product Hackathon	2025
• Built a multi-agent autonomous exam-generation system with a RAG pipeline and Next.js frontend, automating source ingestion, question generation, and education workflows.	
EthosLens — LifeHack 2025	2025
• Built an AI research engine scoring brand ethics and sustainability using FastAPI, Python, and LangChain with automated evidence retrieval.	
DSTA BRINHACK 2025 CTF (top quartile of 400+ teams)	2025
• Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure in a first CTF, finishing in the top quartile among 400+ teams.	
Additional Projects	2023 – 2026
• Project Utopia — real-time situational-awareness dashboard with AI news aggregation and geospatial visualization (2026).	
• Interactive Portfolio Infrastructure — AI-assisted React, TypeScript, and Three.js portfolio with analytics (2026).	
• NVIDIA Disaster Risk Monitoring with Satellite Imagery — geospatial AI with U-Net semantic segmentation on GPU/ASPIRE 2A workflows (2023).	
• Fine-tuning LLMs for Cybersecurity — coursework exploring Mistral/LLaMA fine-tuning with AutoTrain and AutoGen (2024).	

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre	Singapore
<i>Elderly Care Volunteer</i>	Sep 2025 – Present
• Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.	
STEM Inc. (ASRJC)	Singapore
<i>Astronomy Head & Robotics Member</i>	Mar 2019 – Aug 2020
• Led astronomy outreach, competed in the 7th Singapore Astronomical Olympiad, engineered an autonomous robot for the National Robotics Challenge, and received the ASRJC Outstanding Contribution Award.	

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, Bash, C++, C#, Excel VBA, SQL

AI/ML: BERT, DNN, transformers, PyTorch, RAG, AI agents, GPT-4, Gemini, SEALION, LangChain, ONNX, Chroma, local LLM serving (llama.cpp GGUF), vector search, NLP, deep RL (PPO, A2C, DDPG, DQN), model evaluation; vision: camera ISP, super-resolution, image restoration, 3D CV (projective & epipolar geometry, absolute pose, SfM, bundle adjustment, multi-view stereo); HPC: ASPIRE 2A, CUDA, distributed training

Operations Research: CP-SAT, HiGHS linear and quadratic programming, SciPy, hybrid flow-shop scheduling, SimPy, digital-twin simulation, signal processing (Kalman, Welch PSD, change-point, matrix profile), robust optimization, graph optimization, network flow, Dijkstra/SSSP/APSP, decision analytics, statistics, mathematical modeling

Cloud/DevOps: AWS (Fargate, Route 53), Terraform, Azure, Docker, Redis, Kafka, CI/CD, Bitbucket CI/CD, GitHub Actions, Supabase, API design, Codex

Security: Burp Suite, Wireshark, SSRF, CSRF, SQL/NoSQL injection, authentication bypass, OAuth2/JWT, OWASP Top 10, secure design, fuzzing, reverse engineering, custom exploit scripting, cloud security (AWS IMDSv2, metadata APIs)

Web/Full-Stack: React, Next.js, Three.js, Matter.js, FastAPI, Flask, aiohttp, asyncio, GIS mapping, Singpass Login & Myinfo (OIDC), Konva.js

Architecture/Product: Solution architecture, manufacturing systems, stakeholder discovery, rapid prototyping, accessibility, product discovery, user testing, dashboards

Hardware/Embedded: Edge AI / Hailo model deployment, autonomous robotics (sensor integration)

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt); Docker for the Absolute Beginner; Reverse Engineering Windows Executables; Excel 2019 Advanced; Deep Learning CNN and RNN with Python (Packt); Learning C# by Developing Games with Unity (Packt)